The data fields of the class can be obtained by answering this question: What are the properties of the objects of this class? The functions can be obtained by answering this question: What can the objects of this class do?

Next, we'll look at associating many types of objects in relationships.

A.3.2 Classes and relationships

A problem design can be defined by different types of classes in relationships such as

- Inheritance (generalization/specialization)
- Composition
- Association

Other relationships are possible, but these will be useful in the context of smart contracts and blockchain-based decentralized applications.

INHERITANCE

Generalization and specialization of classes, known as *inheritance hierarchy*, is used for representing a hierarchical structure of classes, as shown in figure A.6. Continuing with the automobile example, we can specialize the basic design by using various characteristics and behaviors. In this case, you can see the specialization of the `Auto` class into `Sedan`, `Truck`, and `Van` classes. Here, only one item has been added to the data field of each class (parameters for the number of doors, presence of a cargo bed, and number of